

Mapping Organizational Gradients of Cortical Growth using Spectral Normative Models

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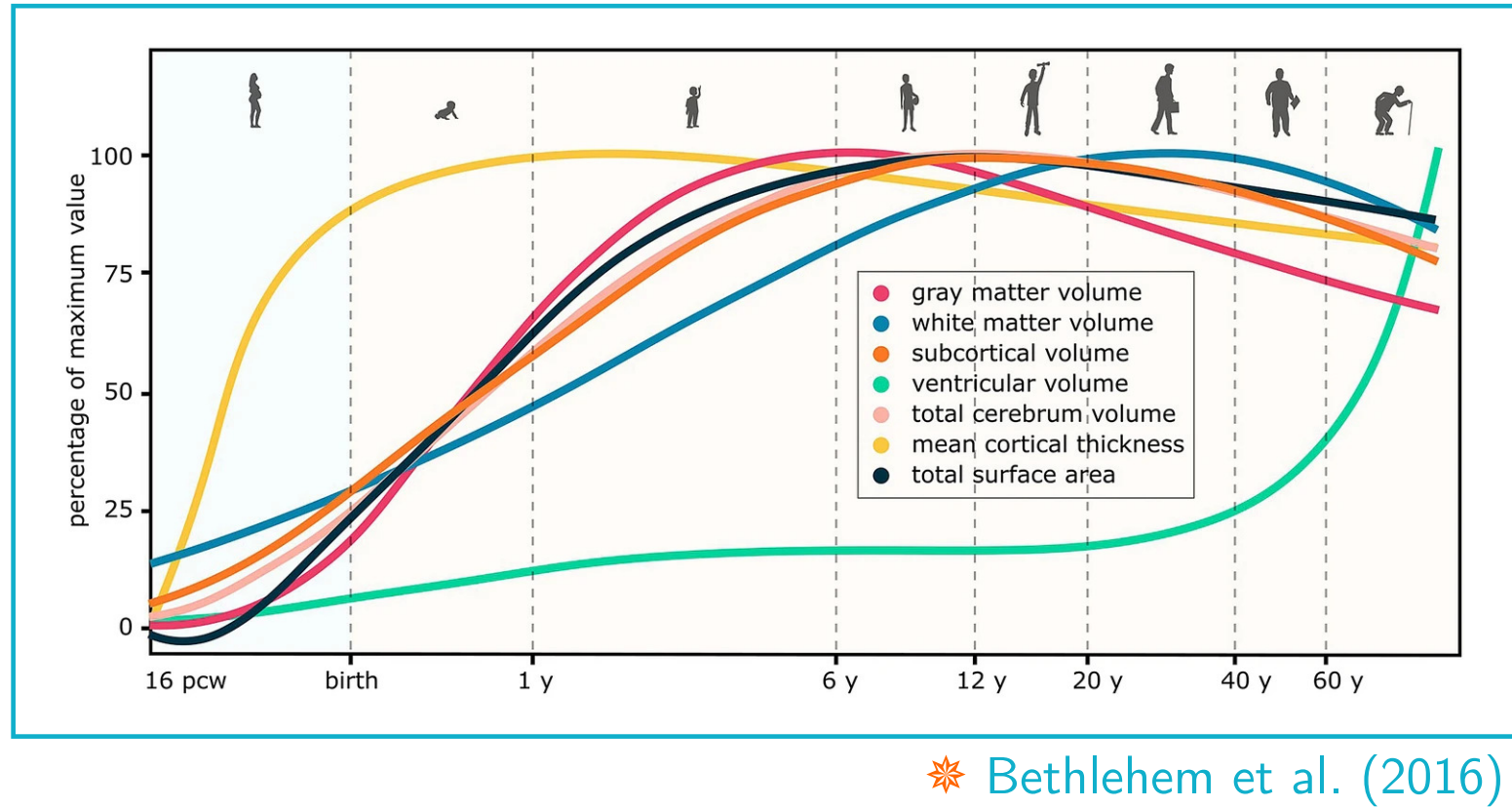
Affiliations are listed in the online version (scan QR code).



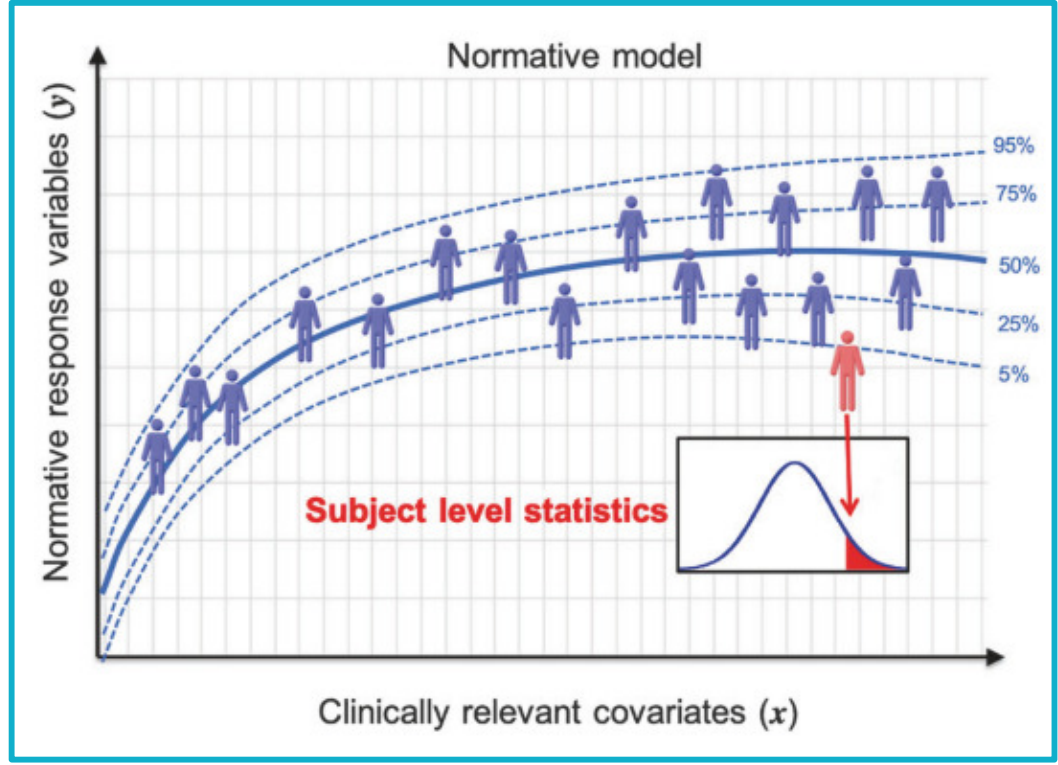
Introduction

• Lifespan brain charts provide a framework for understanding how cortical organization evolves from childhood through aging (Bethlehem et al., 2022).

• Lifelong changes in cortical thickness reflect developmental and aging-related processes (Sowell et al., 2004), yet how these processes are macroscopically organised across the cortex remains poorly understood.



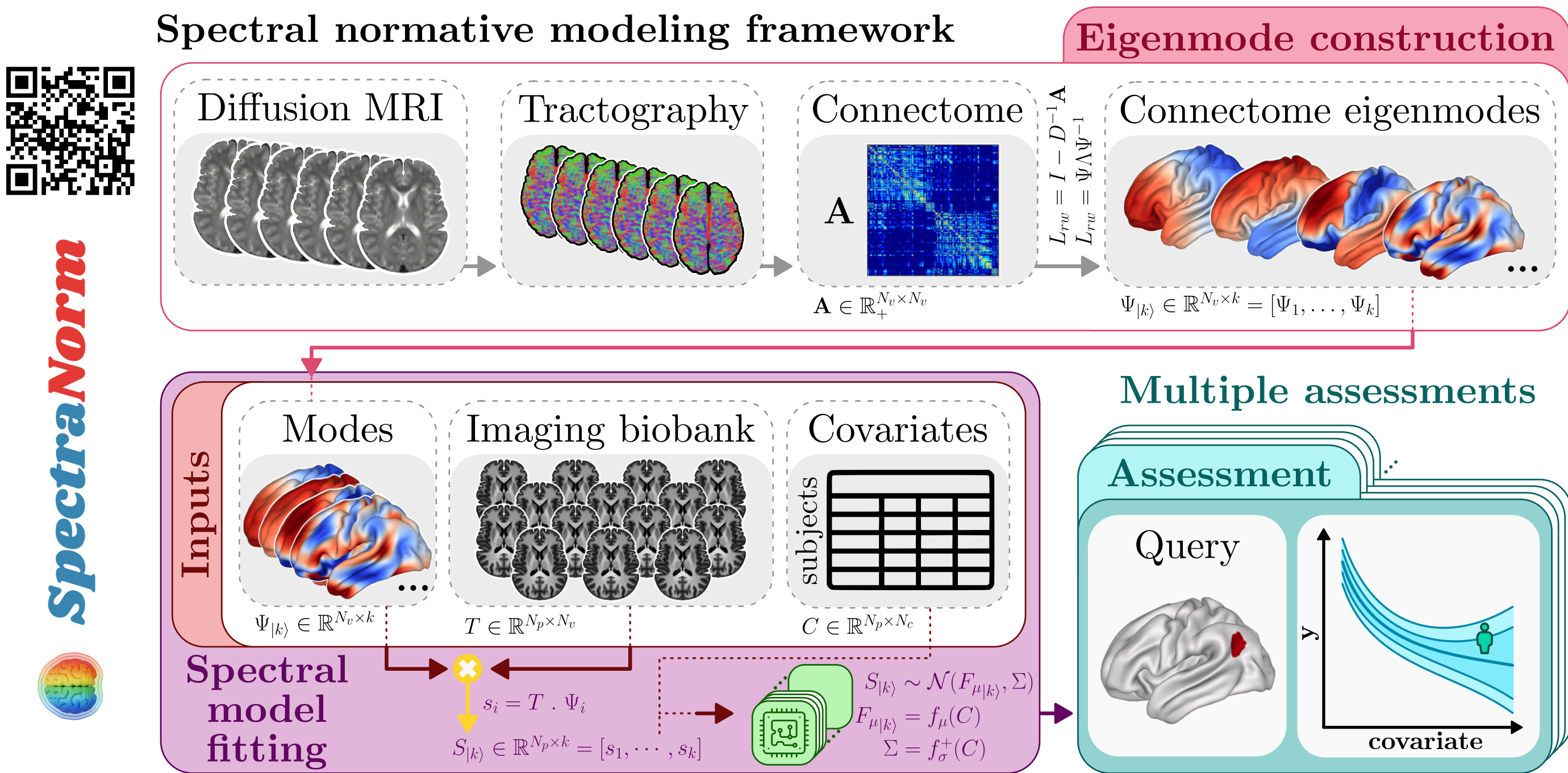
✱ Bethlehem et al. (2016)



✱ Marquand et al. (2019)

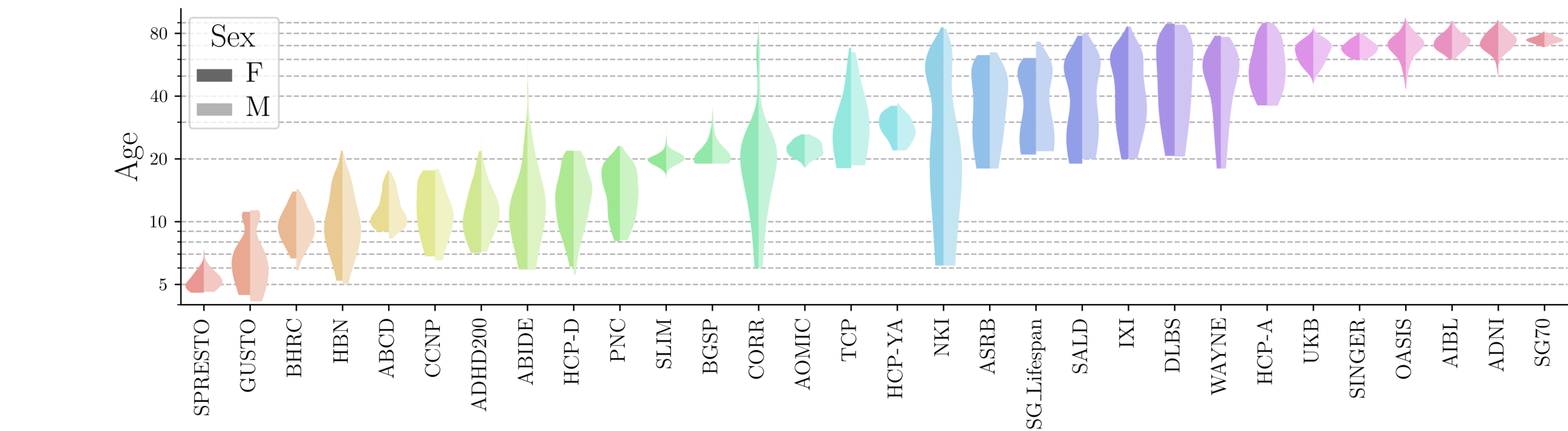
• Normative models have been instrumental in characterizing population-wide distributions of cortical phenotypes within healthy populations (Marquand et al. 2019).

• Building on this foundation, we developed **Spectral Normative Modeling (SNM)**, a flexible framework for efficient normative inference across spatial scales.

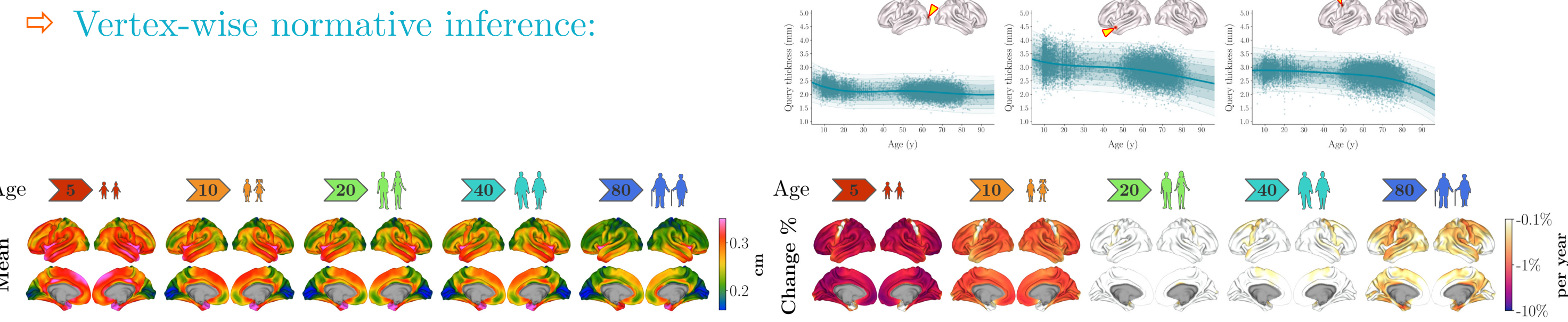


• We utilize this framework to ask whether lifespan cortical thickness trajectories decompose into interpretable, hierarchically organized growth gradients; and if so, what neurobiological processes do they reflect?

Methods

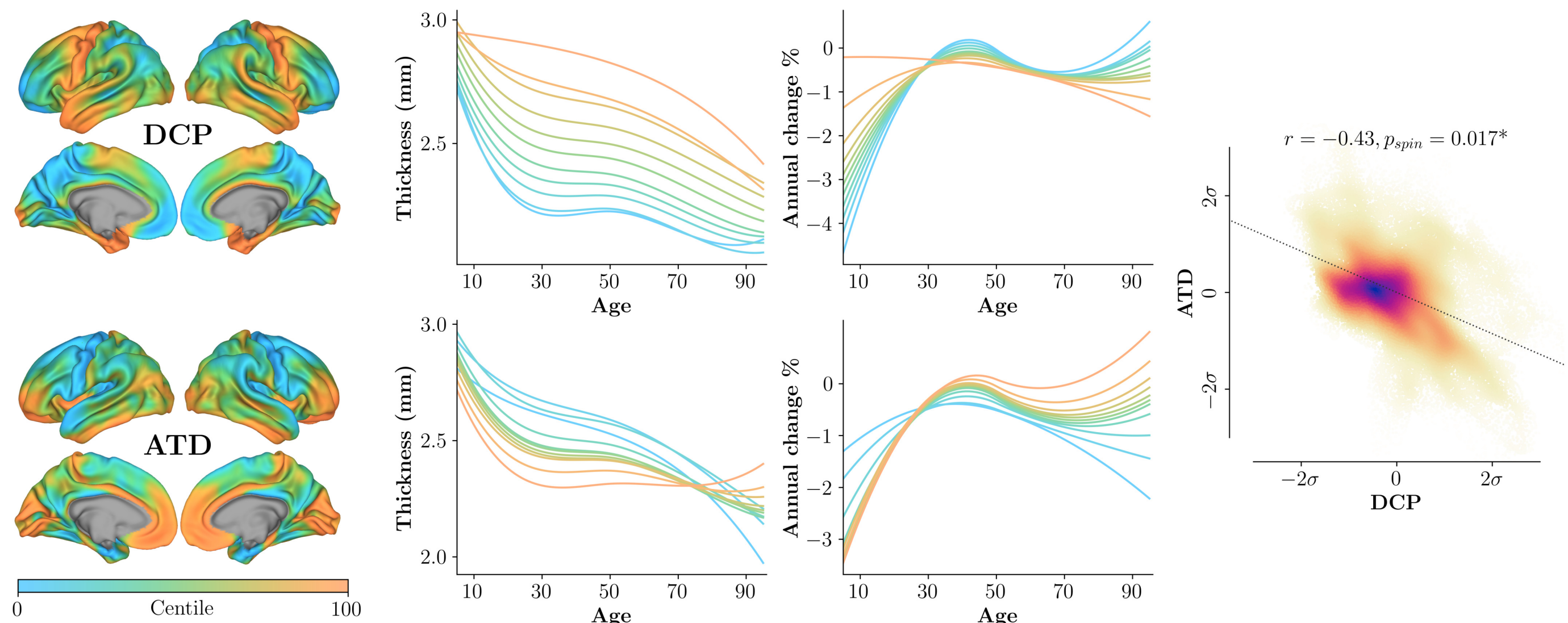


• SpectraNorm was applied to 78,405 scans (ages 5–95, 30 datasets, 189 sites), yielding vertex-resolution lifespan charts of cortical thickness:



Rate-of-change maps were averaged over two age windows:

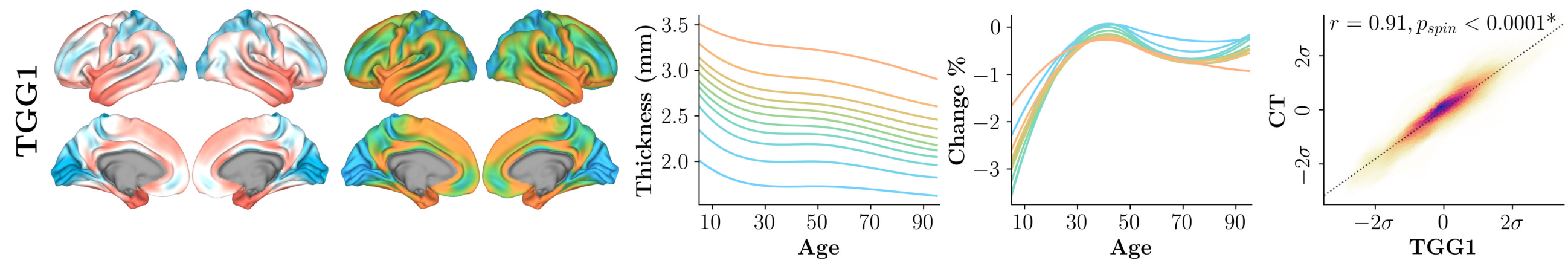
- ➔ **Aging Thickness Decline (ATD):** ages 40–95
- ➔ **Developmental Cortical Pruning (DCP):** ages 5–20



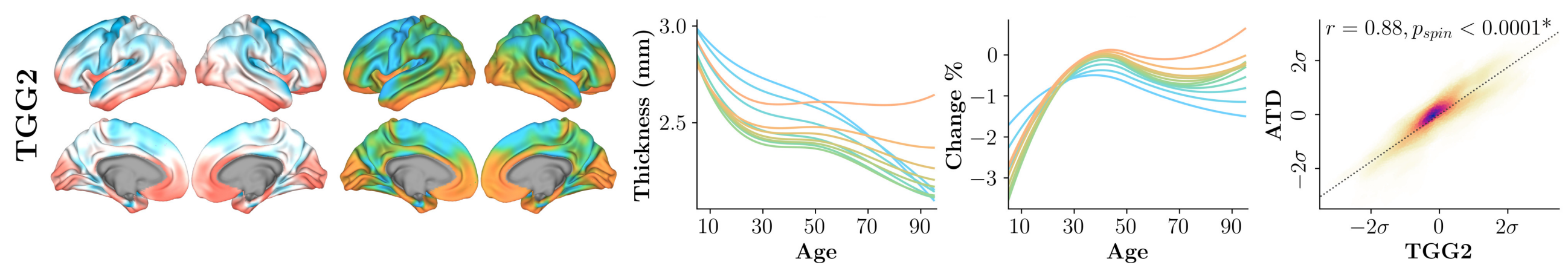
Results

• Principal component analysis of vertex-wise normative trajectories identified **three Thickness Growth Gradients (TGGs 1-3)**, collectively explaining >90% of trajectory variance.

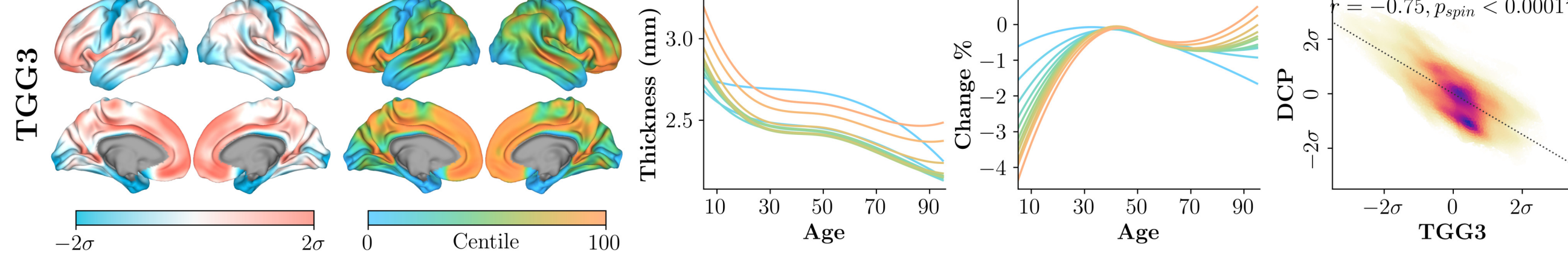
➔ First gradient: **stable global thickness organization across lifespan**



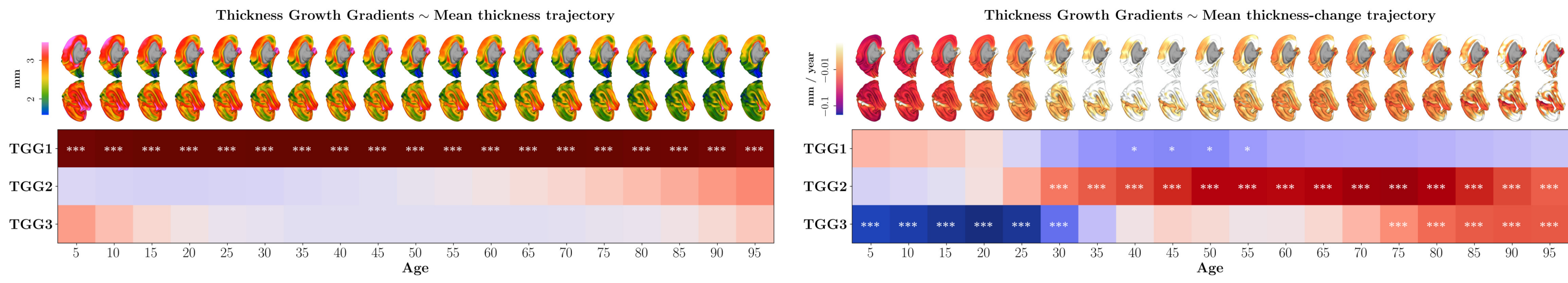
➔ Second gradient: **aging-related cortical thinning**



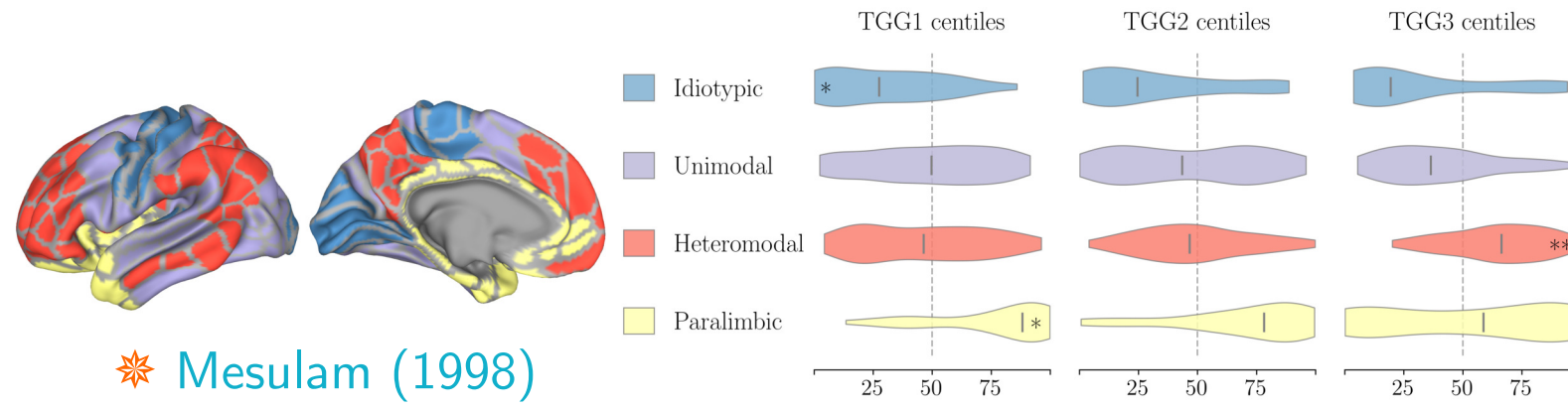
➔ Third gradient: **early-life developmental pruning**



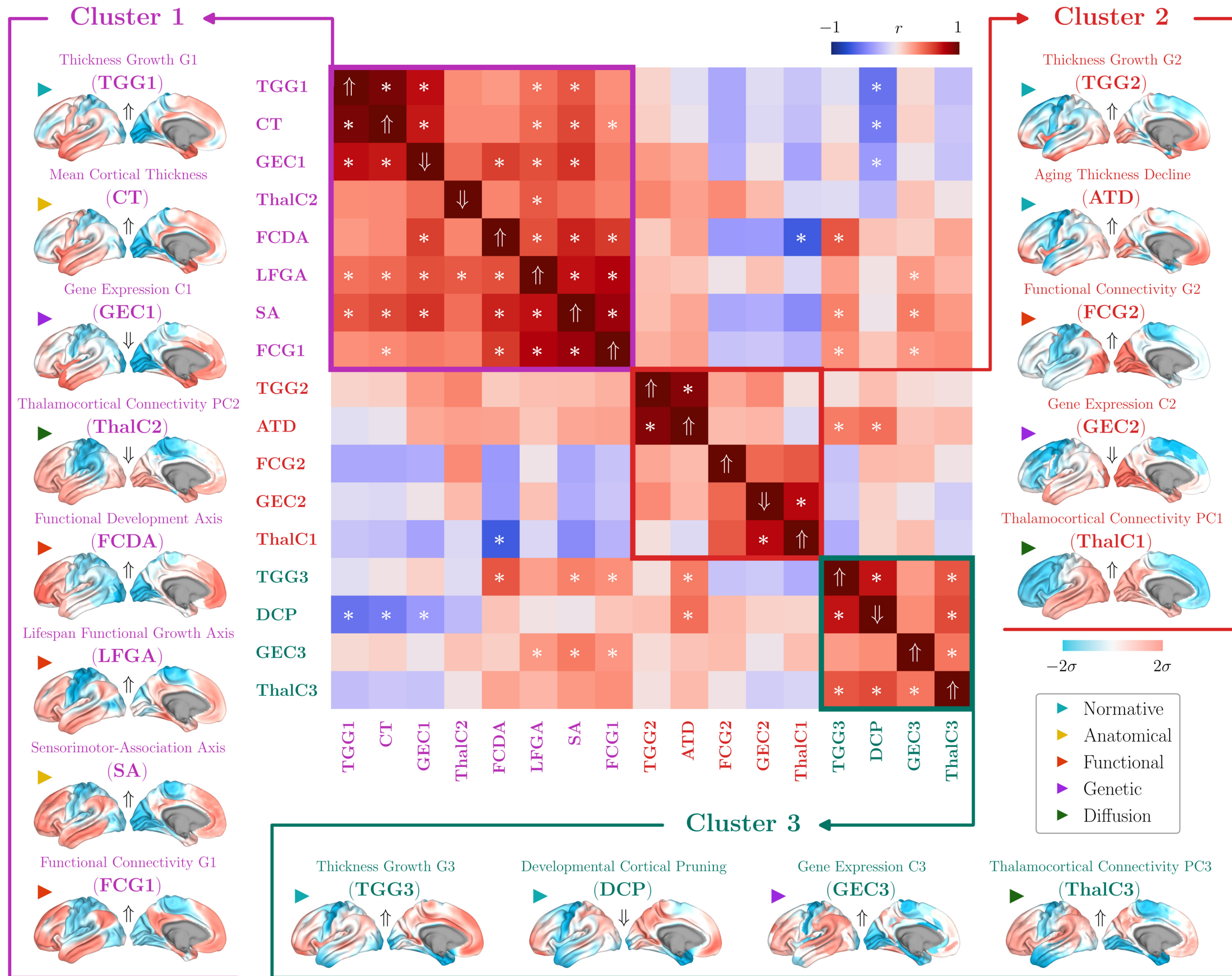
• These gradients were validated against trajectories of thickness mean and rate of change:



• All three gradients align with established laminar cytoarchitectonic hierarchies:



• These gradients were colocated with other previously established cortical hierarchies:



✱ Margulies (2016), Sydnor (2021), Dear (2024), Oldham (2025), Sun (2025)

Concluding remarks

- ✱ We recover three distinct cortical hierarchies, together capturing the majority of lifespan variations in cortical thickness.
- ✱ TGG1 recapitulates stable global thickness organization across the lifespan, aligning with the SA axis and principal transcriptional/functional hierarchies.
- ✱ TGG2 captures aging-related thinning along the dorsoventral axis, clustering with GEC2, a transcriptional component enriched for metabolic processes.
- ✱ TGG3 indexes early-life developmental pruning, grouping with GEC3, reinforcing links to synaptic plasticity.
- ✱ The anticorrelation between DCP and ATD aligns with TGG3, extending the retrogenesis hypothesis: regions relatively spared by developmental pruning show heightened susceptibility to atrophy in later life.